Valid is not explanatory

What a proof about a cognitive model does and does not show, and an audit of CogLean's results

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Valid is not explanatory

- Lean guarantees **validity**: every theorem follows from its premises, with nothing hidden.
- It cannot say whether the premises **already contain** the phenomenon.
- A theorem "the model shows a bowed serial-position curve" is worth little if the model's definition is a bowed schedule.
- So two standards:
  - *formal*: does it follow? (Lean)
  - *explanatory*: how much of the behaviour is in the premises, and how much emerges? (scientific judgment)

Three levels

1. **A phenomenon**: a pattern observed in people (errors bowed across a list).
2. **A predicate**: a precise statement of the pattern, in Lean. It asserts nothing; it is a target.
3. **A theorem**: a mechanism, under stated conditions, produces behaviour satisfying the predicate.

A predicate in CogLib's `Phenomena` is not an axiom. It becomes a result only when a mechanism is proved to satisfy it.

Derived, or conditional?

- A theorem with a phenomenon among its *hypotheses* is a relation between phenomena: "if recency, then . . . ". Valid, useful, but not a prediction of any model.
- It becomes a prediction when a model *discharges* the hypothesis by proving it.
- The catalog now marks this mechanically: **Given**, the laws a theorem assumes.

Theorem	Given	Status
confident errors from laws	determinacy, certainty, stability	conditional
confident misnaming (EPAM)	— (discharged from EPAM's cells)	derived
flat serial curve	certainty, stability	conditional

32 of 525 theorems are conditional on laws.

Explanatory distance

How far is the result from its premises?

Distance	Meaning	Example
Definitional	a restatement	hierarchical network: cost is distance, so a distance effect
Close	the pattern is largely in the mechanism	selective modification's conjunction effect
Derived	follows from premises that don't mention it	the no-carry bug: deletion + impasse + critics
Compositional	needs two or more mechanisms	confident errors: recognition + associations

The question to ask of every result: *how much of the behaviour is in the premises, and how much emerges?*

Case 1: the serial-position curve

Before

- The model was a schedule: item i learned on trial $2\times$ (distance from an end).
- The bow was in the definition. Valid; close to definitional.

Now

- A rule, over *any* anchors: nearer an anchor first (EPAM's M3).
- Proved: nearer is learned earlier, for any anchors.
- Ends as anchors $\Rightarrow$ the serial-position effect.
- A distinctive item as an anchor $\Rightarrow$ the **isolation effect**.

Still close, but one mechanism now answers to two phenomena.

Case 2: the same deficit, two control regimes

Same operators, same missing carry operator; only the control differs.

	no carry needed	a carry needed
Decision cycle with repair	exact	confident wrong (no-carry bug)
Means-ends analysis (GPS)	exact	no answer

- **Compositional**: knowledge $\times$ control regime.
- **Discriminating**: the regimes differ in observable behaviour, with knowledge held constant.
- Proved for every problem.

What the audit found

- **Strongest**: compositional results: confident errors; the arithmetic view (over-extension gone, typicality kept); the two regimes.
- **Derived**: the no-carry bug; GPS's units-first order; the isolation effect; DOA's counting error share; EPAM's oscillation.
- **Close**: serial position; the conjunction effect; DOA's closed forms.
- **Definitional**: the hierarchical network's costs; properties of algorithms (soundness, merges), which are library facts, not explanations.
- **Premises still unexplained**: sharp associations (assumed in every confident-errors result); prior associations; which items are anchors.

A judgment, written down (`notes/EXPLANATORY_AUDIT.md`); its weakest results are the targets.

Conventions, not a checker

- Phenomena are **targets**, not premises.
- A theorem with a law among its hypotheses is **named and flagged** as conditional.
- Docstrings say which premises are **postulates** and which are **conditions**.
- Properties of algorithms are not presented as explanations.
- When a result is close to its definitions, say so, and look for a **second consequence** of the same mechanism.

Why not a checker? "Who checks the checker?" Lean already shows every premise; what remains is judgment, and the right place for it is in the open.

Next

- The audit's weakest results: a process for semantic search, so distance effects are derived, not defined; sharp associations derived from learning; anchors derived from features.
- **A composition designed around this principle**: practising while sleepy. Fatigue raises errors; learning retains some of them; they persist when alert, unless feedback corrects them first. Stated with each premise marked as postulate, condition, or empirical premise awaiting explanation, and with the boundary cases as theorems too.